PORTFOLIO CASE STUDY

Customer Churn Intelligence

End-to-End SQL Analytics, Customer Segmentation and Machine Learning

A business-focused case study that moves from raw customer data to validated insights, risk segmentation, predictive modeling and practical retention actions.

7,043	26.54%	79.41%	0.857
CUSTOMERS	CHURN RATE	CHURN RECALL	ROC-AUC

THE BUSINESS QUESTION

Who is most likely to churn, why are customers leaving, and where should the business focus its retention effort first?

01 | BUSINESS CONTEXT

Executive Summary

Business challenge

Customer churn weakens recurring revenue, reduces customer lifetime value and increases pressure on acquisition. This project was designed to turn historical customer data into a clear retention decision framework.

Core questions

1. **Where is churn concentrated?**
2. **Which customer attributes signal higher risk?**
3. **Why do customers report leaving?**
4. **Can an interpretable model identify at-risk customers?**

Project deliverables

- **SQL foundation**
  Clean, validated analytical data
- **EDA story**
  Clear patterns, charts and context
- **Risk segmentation**
  Actionable customer priority groups
- **Predictive model**
  Independent test-set evaluation
- **Reusable outputs**
  Saved models, probabilities and metadata

End-to-end workflow

1. **PostgreSQL** — Import + validation
2. **SQL** — Cleaning + analysis
3. **Python** — EDA + visuals
4. **Segmentation** — Rule-based priorities
5. **Modeling** — Logistic regression
6. **Action** — Risk scores + recommendations

Technology stack

PostgreSQL · SQL · Python · Pandas · Matplotlib · Scikit-learn · Jupyter · Joblib

Business value

The project does not stop at describing churn. It produces reusable risk scores, an interpretable model, and a practical workflow for prioritizing retention resources.

Data Foundation & Analytical Design

7,043 clean customer records	**33** original variables	**0** duplicate customer IDs	**0** label-value mismatches

SQL data foundation

1. **Raw import**
   Preserve the source data as text
2. **Validation**
   Check counts, duplicates and labels
3. **Clean table**
   Convert types and normalize blanks
4. **Quality audit**
   Review ranges and missing values
5. **Business analysis**
   Create reusable SQL insights

Modeling safeguards

Target variable

churn_value (0 = retained, 1 = churned)

Excluded to prevent leakage

- customer_id - identifier
- churn_label - duplicate target
- churn_score - post-analysis risk field
- churn_reason - known after churn

Training design

Stratified 80/20 split; preprocessing fitted on training data only; class-balanced Logistic Regression evaluated on an independent test set.

Responsible interpretation

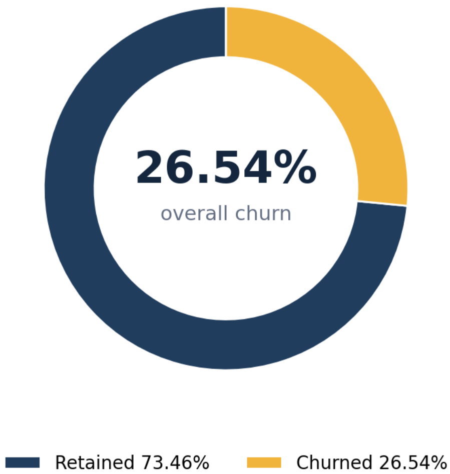
Observed churn differences and model coefficients are treated as associations, not proof of causation. Financial variables are interpreted jointly because tenure, charges, contract type and service usage are closely related.

Reproducibility artifacts

- requirements.txt
- environment_info.txt
- 2 notebooks
- 10 SQL scripts
- 2 saved models
- 3 output files
- model_metadata.json

Where Churn Concentrates

Overall customer outcome



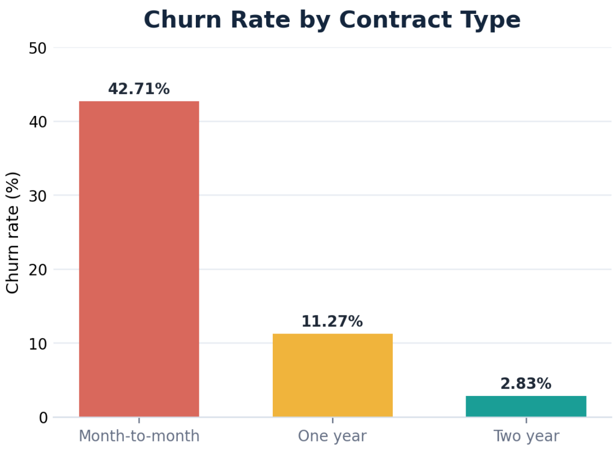
The headline

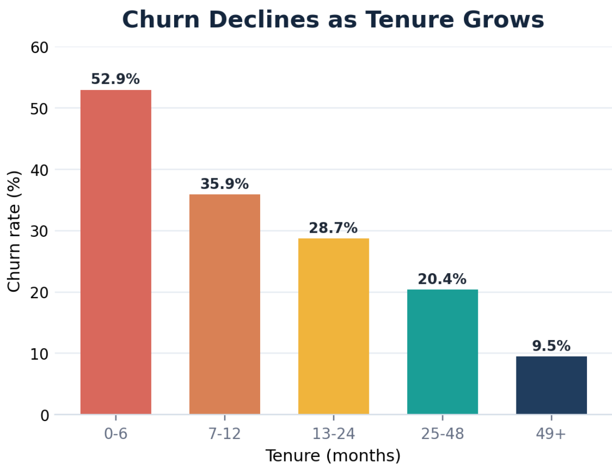
One in four customers churned.

The overall rate is important, but the business opportunity becomes clearer when churn is broken

Primary concentration

88.6% of all churned customers had month-to-month contracts.

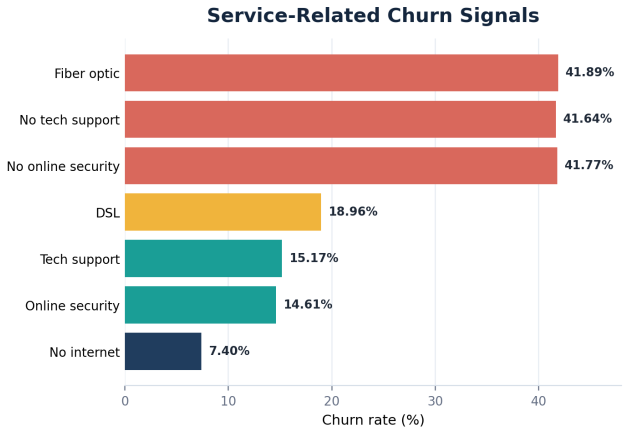


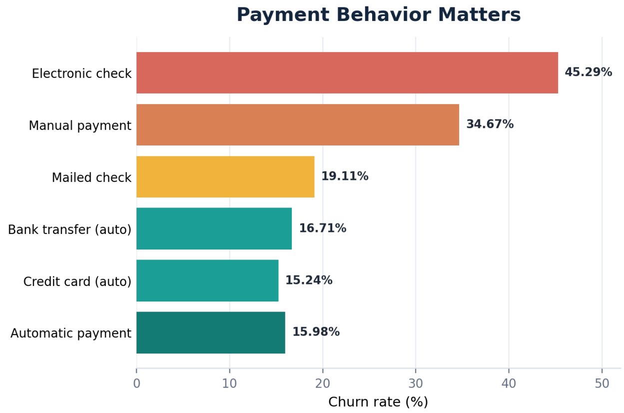


Executive insight

The strongest early signals are low commitment and short tenure: month-to-month customers churned at 42.71%, while customers in their first six months churned at 52.94%.

Operational Signals





Customer profile signals

41.68%
Senior customers
vs 23.61% non-senior

32.55%
No dependents
vs 6.52% with dependents

32.96%
No partner
vs 19.66% with partner

0.76 pp
Gender gap
not a meaningful differentiator

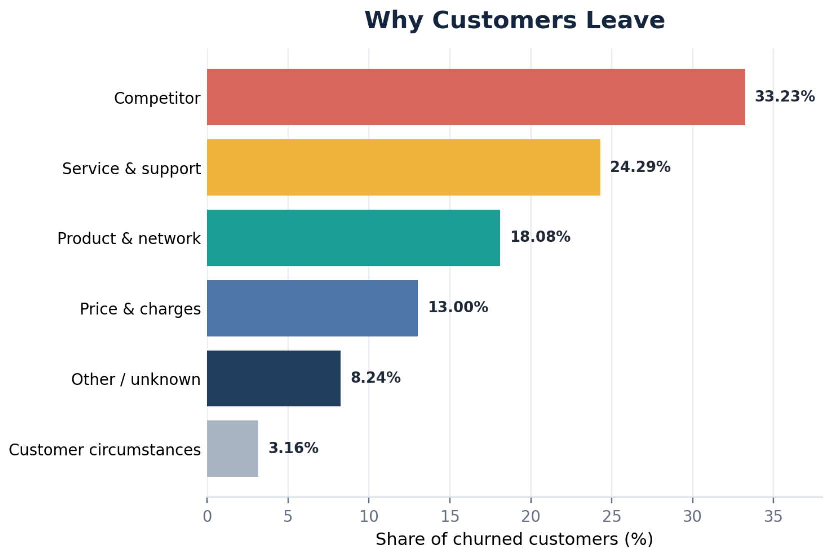
Operational insight

Fiber optic, missing support/security, electronic check and manual payment all show elevated observed churn. These are useful targeting signals, but should be interpreted together rather than as isolated causes.

Retention opportunity

Test service-support bundles and automatic-payment enrollment in the highest-risk customer groups, especially during the first six months.

Why Customers Leave



#1 individual reason

Attitude of support person

192 customers | 10.27% of churn

Support experience was the most frequent individual churn reason.

75.60%

of reported churn

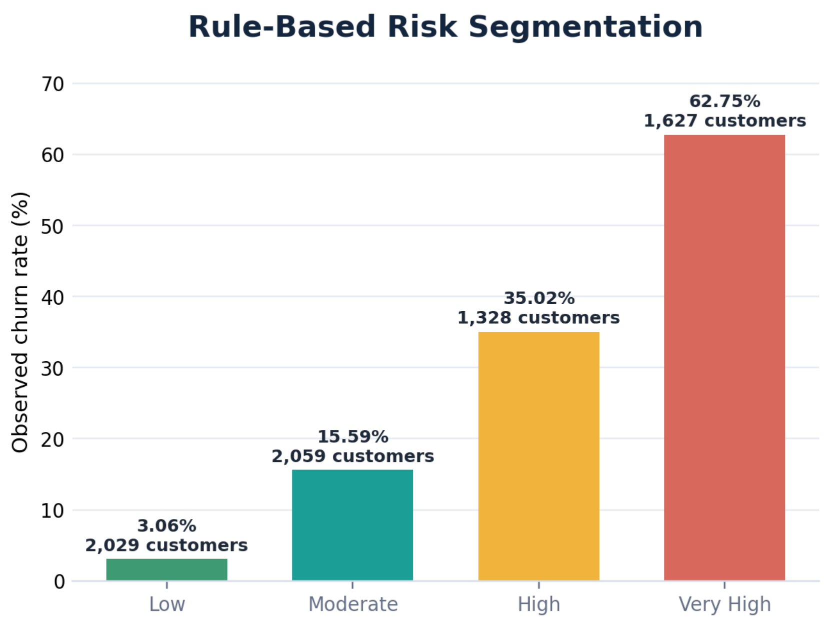
came from competitor, service/support, or product/network categories combined.

Preliminary high-risk customer profile

- Short tenure
- Month-to-month
- Fiber optic
- No tech support
- No online security
- Electronic check
- Manual payment
- Higher service cost

No single trait should determine action. The strongest retention decisions combine risk probability, customer value, service history and the cost of intervention.

From Analysis to Action



Segment summary

Very High 62.75%
1,627 customers

High 35.02%
1,328 customers

Moderate 15.59%
2,059 customers

Low 3.06%
2,029 customers

42% of customers

contained 79.5% of all churn

How to operationalize the segments

1. **Prioritize**
   Start with Very High and High Risk
2. **Add value**
   Combine risk with CLTV and service history
3. **Intervene**
   Assign the lowest-cost effective action
4. **Measure**
   Track uplift, cost and retained value

Important distinction

This rule-based segmentation is an exploratory business tool. It complements - but does not replace - the independently evaluated predictive model.

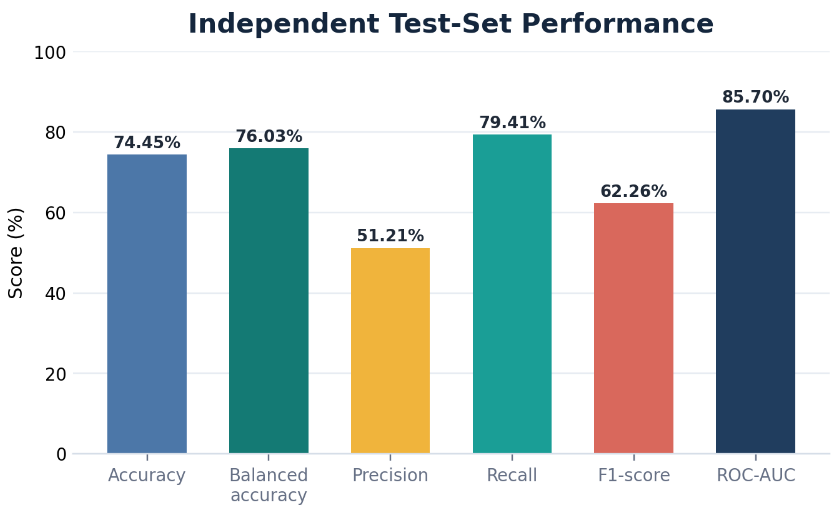
Predictive Model Performance

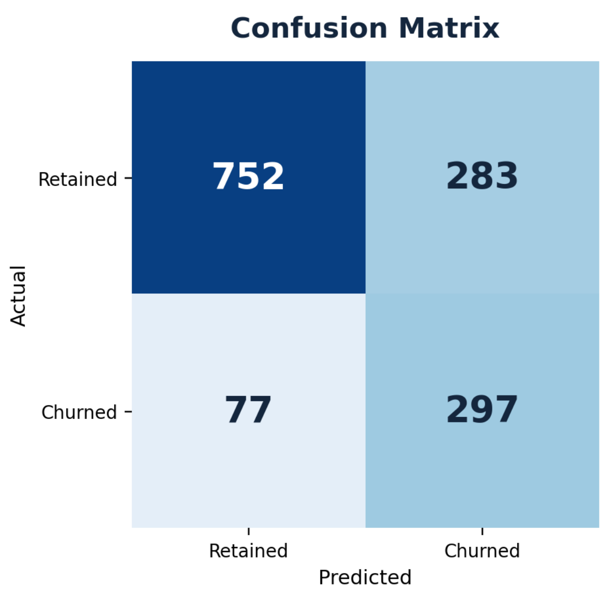
80/20
stratified split

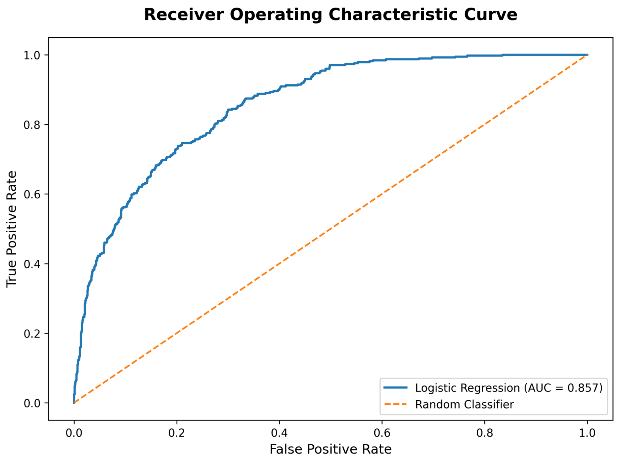
4 + 16
numeric + categorical

Pipeline
impute + scale + encode

Balanced
Logistic Regression





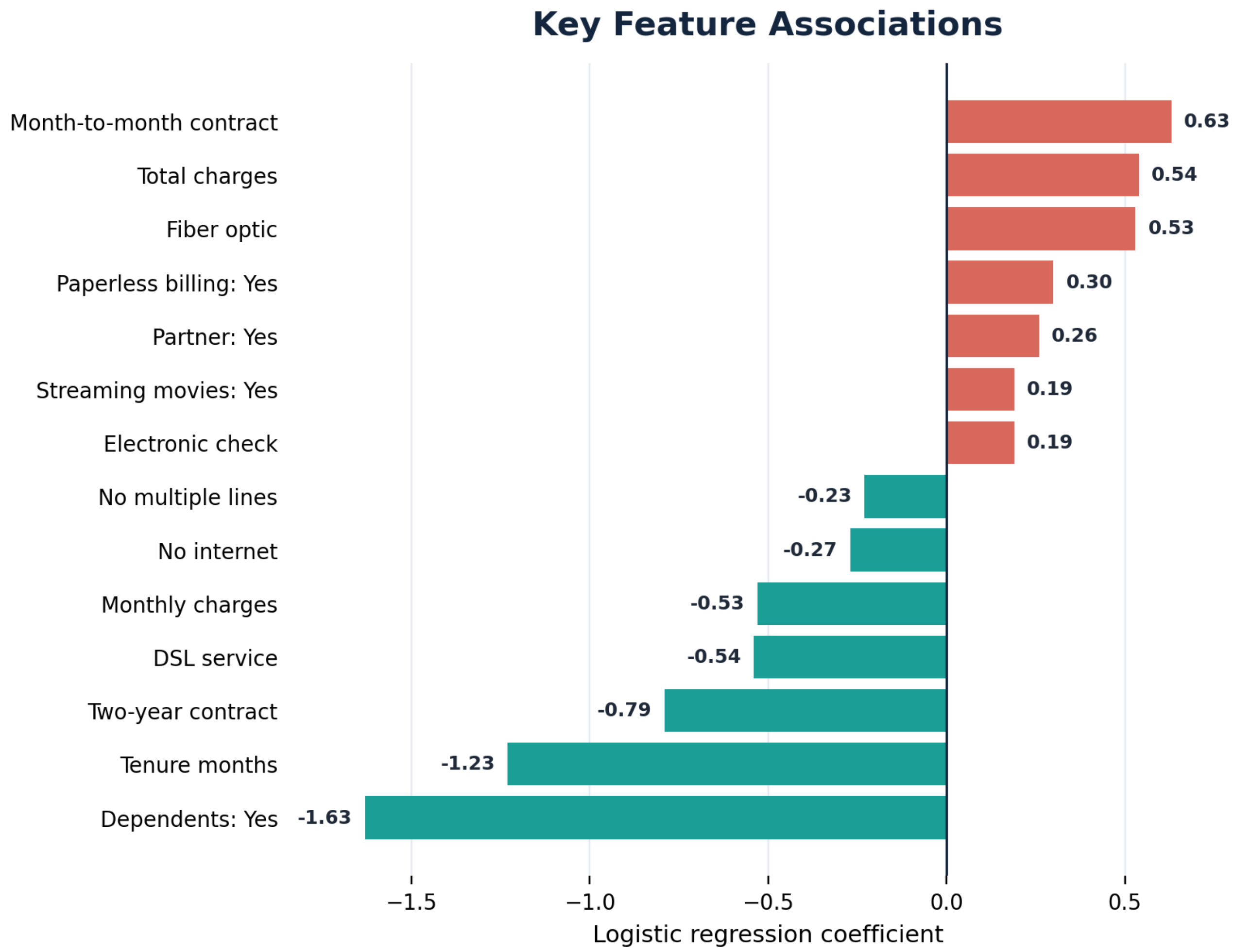


ROC-AUC = 0.857

Interpretation

Recall reached 79.41%. The 283 false positives mean the operating threshold should reflect retention cost and customer value.

What Drives Predicted Risk



Higher predicted risk

Month-to-month contracts
Fiber optic service
Paperless billing
Electronic check payment

Lower predicted risk

Longer tenure
Two-year contracts
Having dependents
DSL / no internet service

Interpretation note: coefficients are conditional associations after accounting for the other modeled variables. They are not causal effects.

Business Recommendations & Next Steps

1 Protect the first six months

Structured onboarding, satisfaction checks and proactive support.

2 Convert suitable monthly contracts

Test annual incentives, loyalty benefits and bundled value.

3 Encourage automatic payment

Simplify enrollment and test small incentives for adoption.

4 Build a risk-led workflow

Prioritize by churn probability, CLTV and intervention cost.

5 Review fiber optic experience

Investigate reliability, pricing, expectations and competition.

6 Strengthen support & security

Test bundled tech support, online security and education.

7 Improve service interactions

Train staff, monitor response time and strengthen resolution.

8 Monitor competitors continuously

Benchmark speed, data, devices, promotions and contracts.

EXPECTED BUSINESS VALUE

Faster prioritization | More proactive retention | Better use of intervention budget | Reusable customer risk scores

Project artifacts

10 SQL scripts | 2 notebooks | 15 visuals | 2 models | 3 outputs

Next validation steps

Cross-validation | calibration | threshold optimization | live testing

"I turn raw data into clear, business-focused decisions."